# Mayur More


M.Tech., Mechanical Engineering
Specialization: Thermal and Fluids Engineering
Indian Institute of Technology Bombay

**DOB:** 30/12/2000
**Gender:** Male


## EDUCATION

| Examination | University | Institute | Year | CPI / % |
|---|---|---|---|---|
| Post Graduation | IIT Bombay | IIT Bombay | 2025-27 | 8.55 |
| Graduation(Mechanical) | Shivaji University, Kolhapur | Government College of Engineering, Karad. | 2022 | 9.13 |
| Intermediate | Maharashtra State Board | Ligade Patil Jr college of science, Karad. | 2018 | 89.56% |
| Matriculation | Maharashtra State Board | K.D.S Highschool, patan | 2016 | 96.00% |

## M.TECH PROJECT & SEMINAR

**Aerodynamic and Thermodynamic Design of a Multi-Stage Axial Flow Compressor** (May'26-Present)
*(M.Tech Project — Guide: Prof. Neeraj Kumbhakarna)*

- **Current work:** Applying a direct 3D blade design methodology utilizing radial equilibrium equations and empirical cascade correlations (e.g., Carter's and Howell's rules) to translate 1D mean-line data into full three-dimensional blade coordinates for 19 compressor stages.
- Exploring alternative vortex strategies, such as constant reaction and forced vortex designs, to achieve favorable radial distributions of blade loading.
- **Future work:** Conduct high-fidelity Computational Fluid Dynamics (CFD) analysis to verify stage performance and investigate complex flow phenomena like tip leakage and secondary flows.
- Integrate the developed off-design performance map with the Part Electric Gas Turbine (PEGT) control logic to optimize variable-speed motor operation.

**Aerodynamic and Thermodynamic Design of a Multi-Stage Axial Flow Compressor** (Jan'26-May'26)
*(M.Tech Seminar — Guide: Prof. Neeraj Kumbhakarna)*

- Designed a 19-stage axial flow compressor for a marine Part Electric Gas Turbine, targeting an overall pressure ratio of 10.06 and a mass flow rate of 20 kg/s.
- Developed a dynamic 1D stage-by-stage solver in MATLAB to compute velocity triangles and evaluate De Haller diffusion limits based on primary turbomachinery textbook principles.
- Generated a complete off-design performance map ensuring speed lines follow natural aerodynamic curves, validating the peak isentropic efficiency of 83-84%.

## INTERNSHIPS

**CFD Intern — Vedaero** (Nov'25-Feb'26)
*"Flow Over a Flat Plate (Boundary Layer Development)"*

- Acquired foundational knowledge of Computational Fluid Dynamics (CFD) principles and developed practical proficiency in problem setup, solving, and post-processing using ANSYS Fluent.
- Simulated the development of a laminar boundary layer over a 1 m flat plate within a 2 m $\times$ 0.5 m computational domain subjected to a uniform inlet velocity of 5 m/s.
- Analyzed the velocity boundary layer growth, mapped the wall shear stress distribution, and calculated the skin friction coefficient.
- Validated the MATLAB simulation results against the theoretical Blasius analytical solution to accurately determine friction drag.

## COURSE PROJECTS

**Thermal Management of Hydrogen Fuel Cell** (Jan'26-Apr'26)
*(ME 770 - Thermal Design of Electronic Equipment)*

- Designed 3D biomimetic (bio-leaf) and wavy flow channel configurations to maximize convective surface area and enhance heat extraction from the PEM.
- Conducted CFD and heat transfer simulations to analyze steady-state uniform heat generation, aiming to stabilize the membrane temperature at the target 333K.
- Demonstrated that complex flow geometries significantly improve thermal bridging compared to standard serpentine channels under 50% oxygen consumption conditions.

**Free Stream Flow Across a Square Cylinder** (Jan'26-Apr'26)
*(ME 415 - Computational Fluid Dynamics and Heat Transfer)*

- Developed a 2D incompressible flow solver using a semi-explicit pressure correction approach on a $220 \times 176$ non-uniform staggered grid.
- Implemented the Finite Volume Method (FVM) with a flux-based methodology, utilizing First Order Upwind schemes for advection and Central Differencing for diffusion terms.
- Simulated free stream flow at a Reynolds number of 100 to capture and analyze aerodynamic wake characteristics and vortex shedding.

**Prediction of Warpage and its Control Mechanisms in Panel-Level Advanced Packaging** (Jan'26-Apr'26)
*(ME 768)*

- Investigated warpage in Fan-Out Panel-Level Packaging (FOPLP) caused by Coefficient of Thermal Expansion (CTE) mismatch and polymer curing shrinkage during cooling (230°C to room temperature).
- Evaluated severe structural deformations (exceeding 40 mm) on large-scale Gen-3 panels ($550 \times 650$ mm) and analyzed consequences like handling issues and alignment failures.
- Proposed strategic control mechanisms to mitigate structural strain, addressing the primary manufacturing barriers to FOPLP industrialization.

## BTECH PROJECT

**Design and Fabrication of Pico Hydroelectric Turbine**

- Developed a decentralized renewable energy solution to reduce reliance on centralized power systems.
- Led the fabrication process and ensured quality control throughout the project.
- Enhanced teamwork, problem-solving, and renewable energy expertise.

## INDUSTRIAL TRAINING

**Total Quality Management (TQM) & Six Sigma Techniques**

- Completed a 15-day comprehensive training during the 3rd year of engineering studies.
- Developed skills in process optimization, defect reduction, and efficiency enhancement.
- Gained proficiency in data-driven decision-making and statistical analysis.

## TECHNICAL SKILLS

- **Programming & Scripting Languages :** Matlab
- **Tools & Libraries :** MATLAB, Ansys, LATEX, Fusion 360, Solidworks

## KEY COURSES

- Computational Fluid Dynamics and heat transfer • Fluid Dynamics
- Advanced Heat Transfer • Thermal Design of Electronic Equipments
- Microsystem Packaging • Combustion in Automobile and Gas Turbine
- Computational Methods in Thermal and Fluid Engineering •

## CERTIFICATIONS

- **MATLAB® Course — IIT Madras Pravartak.** Successfully completed training to analyze data, develop algorithms, and model systems using MATLAB.
- **Siemens NX Training — Internshala.** Completed comprehensive training focused on 3D modeling and design, featuring a specialized practical project on airplane wing design.

## SCHOLISTIC ACHIEVEMENT

- Scored 98.431%ile in GATE 2025 ME paper among 62015 students.

## POSITION OF RESPONSIBILITIES & EXTRA-CURRICULARS

- **Teaching Assistant for the course, MS101 : Maker's Space** (Aug'25-Dec'25 & Jan'26-Apr'26)
- Actively participated in the **Inter-Departmental Badminton (Singles)** tournament.
- Competed in the **District-Level Speech Competition**.
- **Hobbies :** Listening Music, Playing Badminton and Cricket, making 2D and 3D animation.